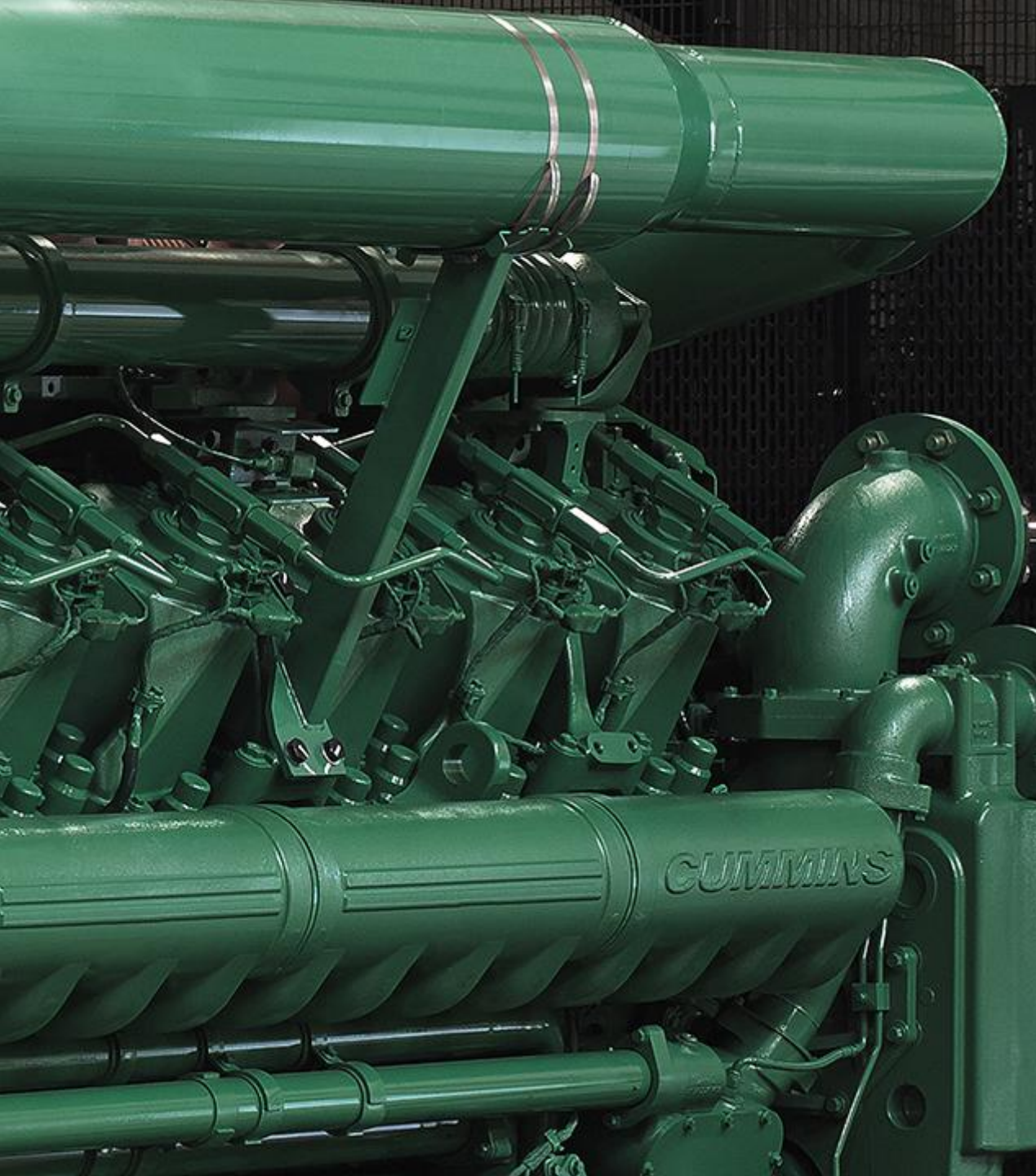




Generator Ratings & Performance Standards

Consultant Development Team

Contents

- ISO 8528-1 Definitions
- ISO Ratings
(COP, PRP, ESP, LTP DCP)
- Derate Factors
- Performance Classes
- Transient definitions
- BMEP
- ISO Load Stages

Industry Standard for Generator Set Ratings

ISO8528



- Is the Internationally Recognised Standard for *Reciprocating Internal Combustion Engine Driven Alternating Current Generating Sets*.
- It applies to both compression ignition (Diesel) and spark ignition (Gas) engines.

Standard Reference Conditions

- Barometric pressure = 100 kPa
- Ambient Temperature = 25°C
- Relative Humidity = 30%

ISO8528-1 Power Rating Definitions

- ISO 8528-1: Defines application, ratings and performance of generator sets.
 - Power Ratings:
 - Emergency Standby Power (**ESP**)
 - Prime Rated Power (**PRP**)
 - Limited Time Prime Power (**LTP**)
 - Continuous Operating Power (**COP**)
 - Data Centre Power (**DCP**)



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- “The power of the generating set is the power output available for consumer loads **at the generating set terminals** excluding the electrical power absorbed by the essential independent auxiliaries.”



ISO8528-1 Definitions

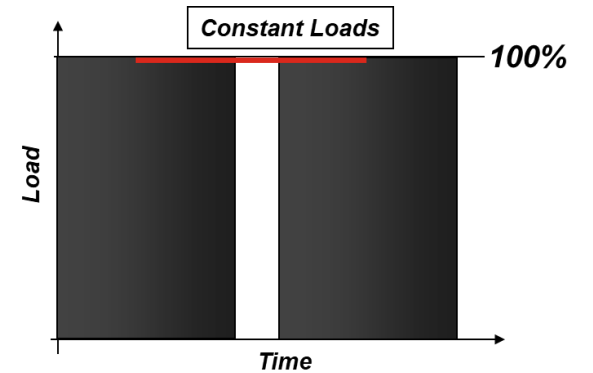
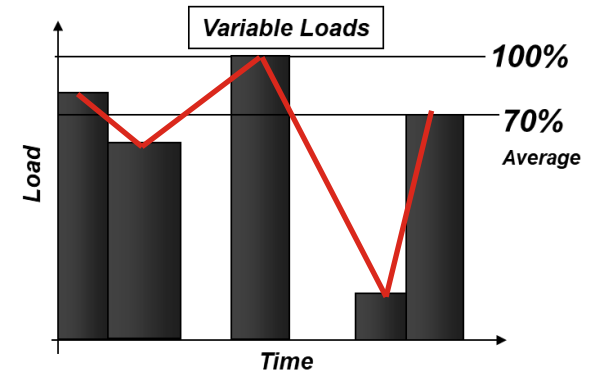
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- “The power of the generating set is the power output available for consumer loads **at the generating set terminals** excluding the electrical power absorbed by the essential independent auxiliaries.”
- **For Clarity, Include ISO 8528-1 Power Rating Definitions in any specifications.**



ISO 8528-1 Ratings

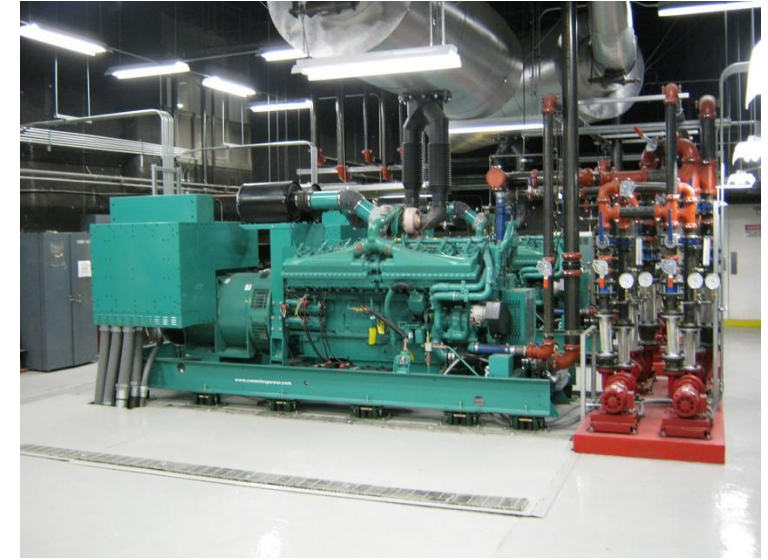
Overview...

Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

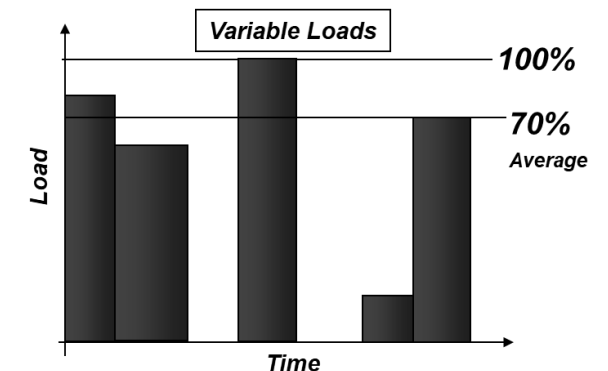


Emergency Standby Power (ESP)

- “Maximum power available during a variable electrical power sequence...for up to **200 h** of operation per year”
- “The permissible average power output over **24 h** of operation shall not exceed **70% of the ESP** unless otherwise agreed by the RIC engine manufacturer”
- Examples for ESP applications include life safety, legally required, or critical loads
- Not required by the Authority Having Jurisdiction (AHJ), but desired to minimize economic losses or equipment damages at a site due to utility power interruptions



Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

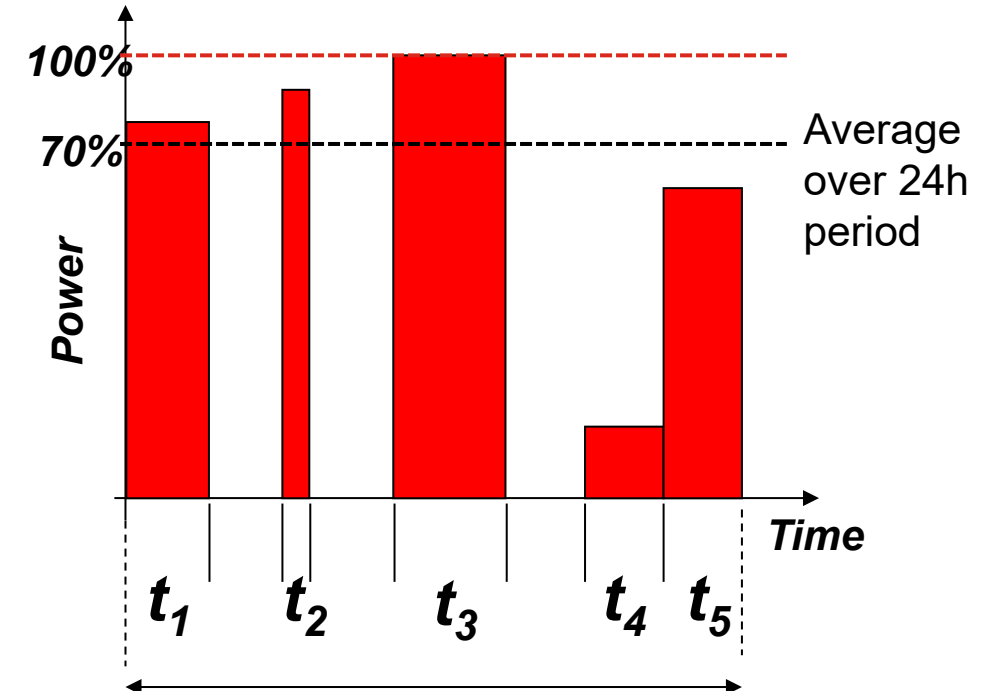


Emergency Standby Power (ESP)

- Maximum power available from any given generator.
- Variable electrical power profile... for up to **200 h** of operation per year”
- The permissible average power output over **24 h** of operation shall not exceed **70% of the ESP** unless otherwise agreed by the RIC engine manufacturer”
- No extended utility paralleling (short term/soft return permitted)
- Examples for ESP applications include life safety, legally required, or critical loads

Limited number of hours in variable load

- Not to exceed 200h / year in variable load profile
- Not to exceed 70% avg. load factor over 24h period
- Avg. does not include periods of shutdown
- No overload capability
- **[Cummins] 500 hours / year**



Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
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Cummins Application



3 units C2500 D5A for black start of a 300MW gas turbine plant in Algeria

Location: Ain Beida, Algeria

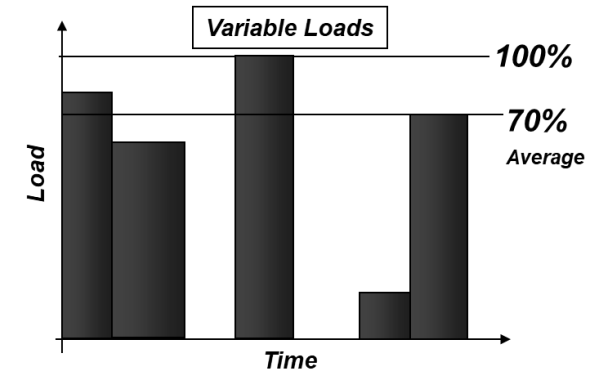
Altitude: 950m ASL

Temperature range: 7°C to 45°C (17°C yearly average)

Humidity range: 41% to 83% (62% average)

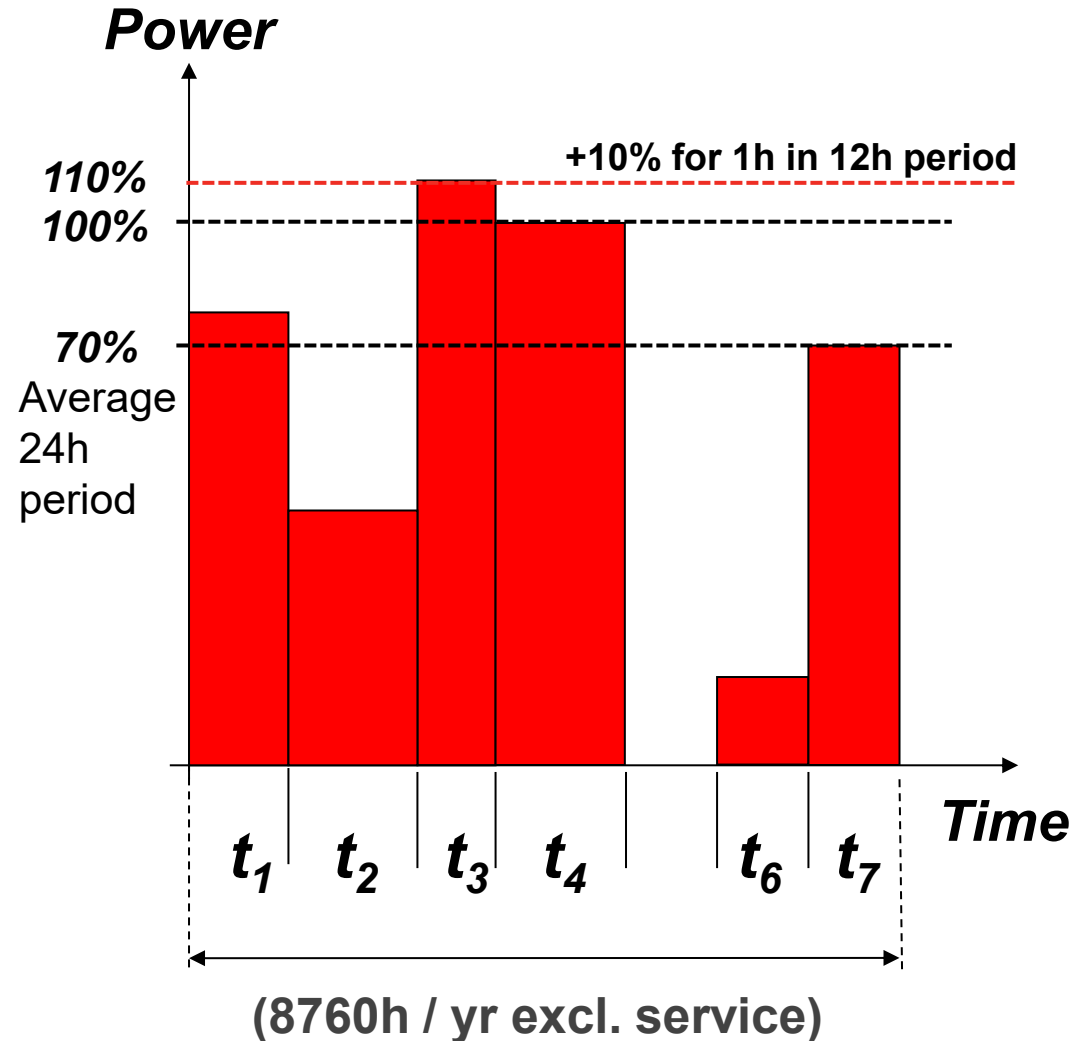
Prime Rated Power (PRP)

- “Maximum power which a generating set is capable of delivering continuously while supplying a **variable electrical load** when operating for an **unlimited number of hours** per year”
- “Permissible **average power output over 24 h** of operation shall not exceed **70% of the PRP**”
- Examples include applications that use on-site generation in lieu of a utility electricity supply, typically where utility power is not available.
- Peak shaving and rate (Extended Utility Parallel) curtailment programs



Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
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Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

Prime Power (PRP)

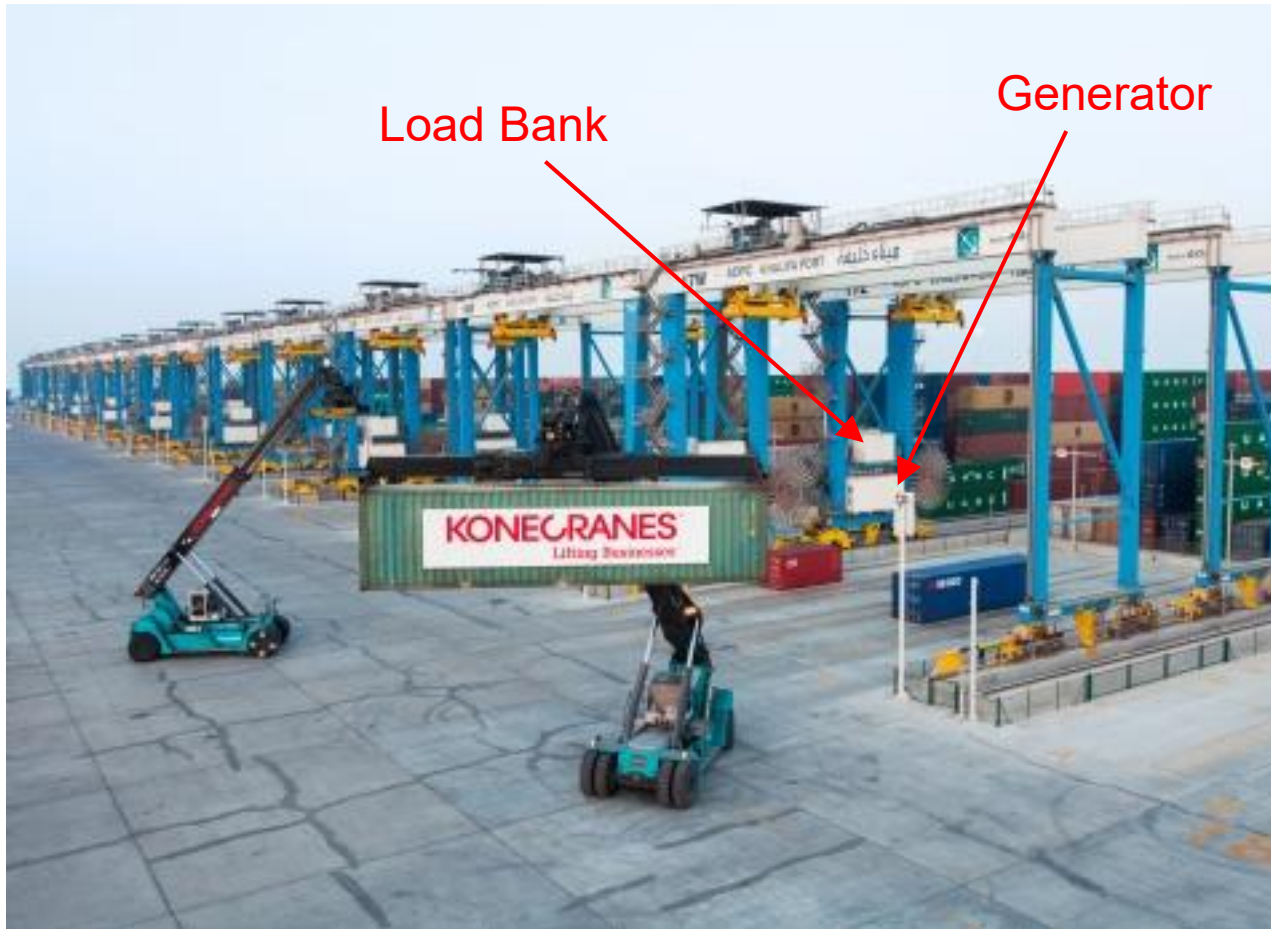


ISO8528-1:

Unlimited number of hours per year in a variable load application

- Not to exceed 70% avg. load factor over 24h period
- Avg. will count loads less than 30% as 30%
- Avg. does not include periods of shutdown
- [Cummins] 10% overload which shall not exceed 25h / year

Cummins Application



Konecranes – Nigeria

500kVA generator powers the dockside gantry cranes.

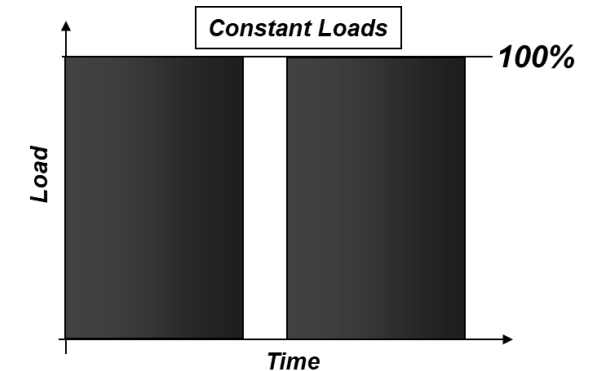
Cranes lift capacity is 40 tonne and can move at 135 metres per minute.

Regenerative loads – max. 10%.

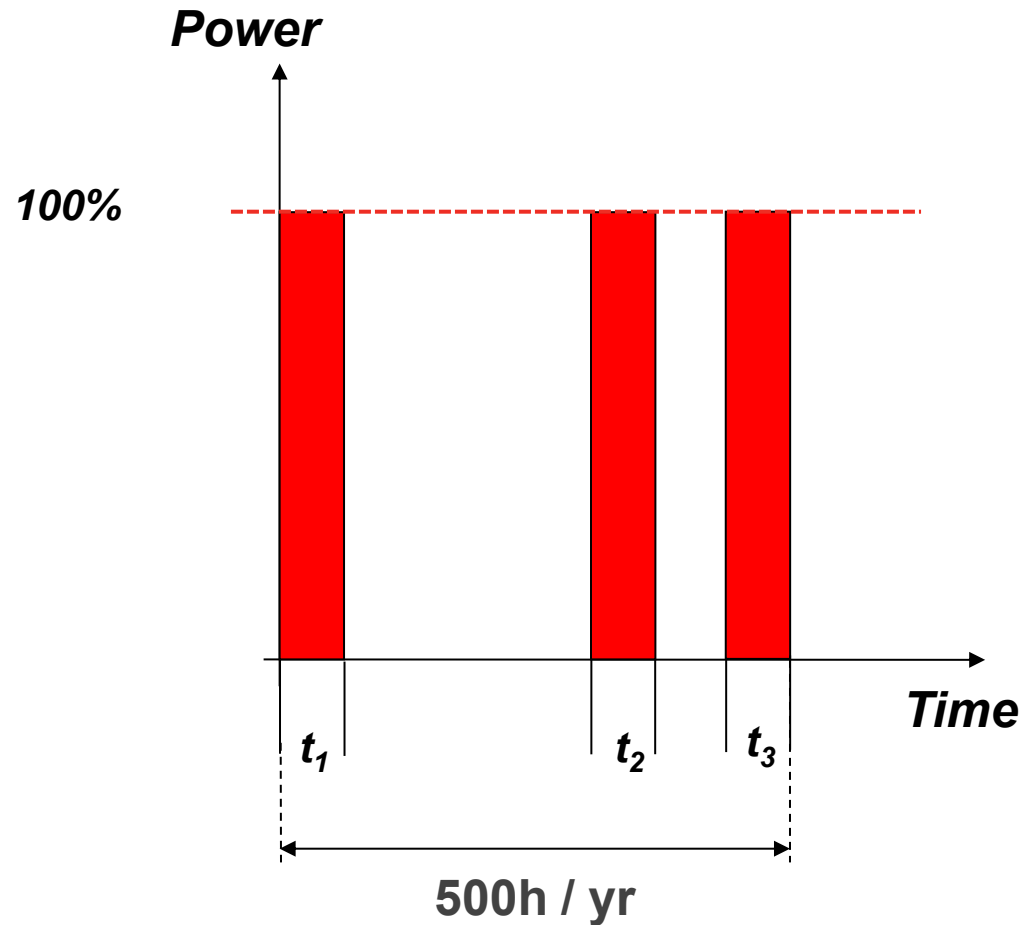
Limited Time Prime (LTP)

- “Maximum power available, under the agreed conditions, for which the generating set is capable of delivering for up to **500 h** of operation per year”
- Examples of LTP applications include base loading and rate curtailment programs

Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited



Limited Time Running Power (LTP)



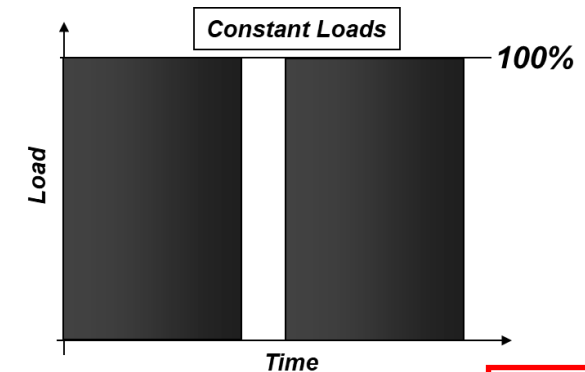
ISO8528-1:

Limited number of hours in a non-variable load application

- Not to exceed 500h / year in constant load profile
- No overload capability
- [Cummins] Engines operated in parallel with utility can be operated for up to 750h / year

Continuous Operation Power (COP)

- “Maximum power which the generating set is capable of delivering continuously while supplying a **constant electrical load** when operated for an **unlimited number of hours** per year”
- Examples of COP applications include paralleling to the utility, base loading, or combined heat and power (CHP)



The Cummins Power Generation lean-burn generator set produces up to 1.75 MW of electricity and 4,000 pounds of steam per hour in a CHP application.

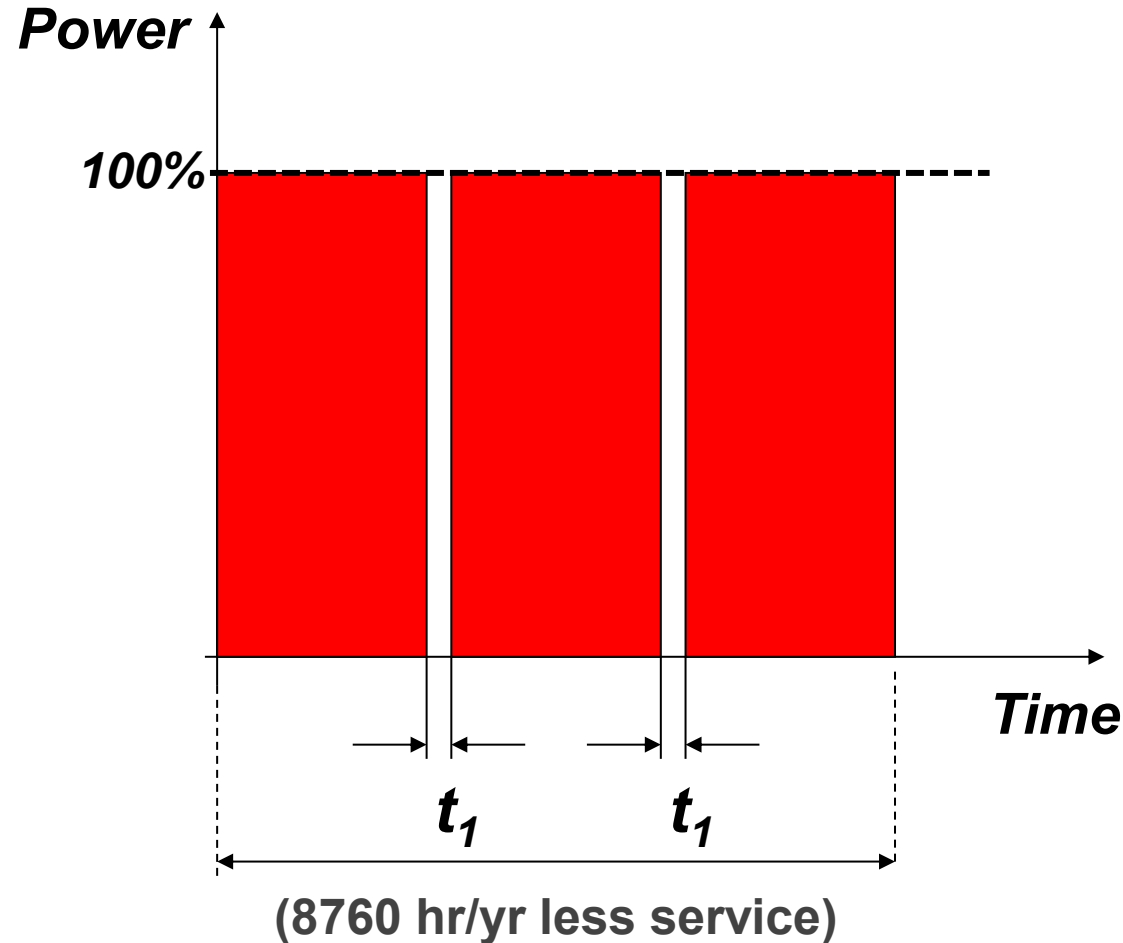


The lean-burn engine's radiator and critical exhaust silencer are roof mounted.

Combined heat and power - Chicago Museum of Science and Industry

Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

Continuous Operating Power (COP)



ISO8528-1:

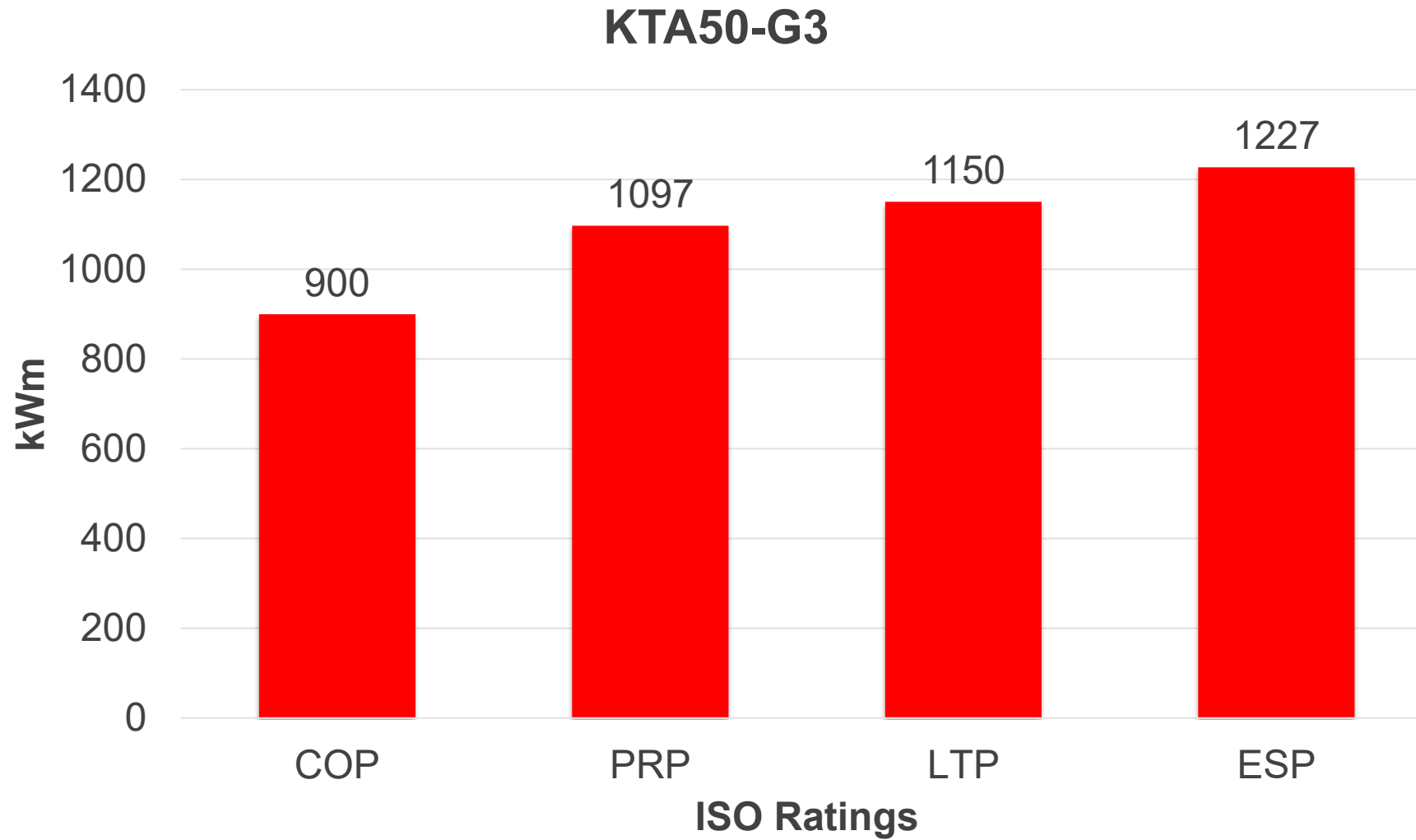
Unlimited number of hours in a non-variable load application

- Constant load profile
- No overload capability

Cummins Application



Comparison



Data Centre Continuous (DCC)

Data Centre Power (DCP)

DCC rating is Cummins own rating for Gensets uniquely for Data Center Applications and now adopted by ISO8528 – 1 : 2018

This rating was developed with the Uptime institute and all Cummins DCC rated gensets are Uptime compliant.

*Maximum power which the generator is **capable of delivering continuously** to a **constant or varying** electrical load for **unlimited hours** in a data centre application.*



Data Centre Power (DCP)

- “Maximum power which a generating set is capable of delivering while supplying a **variable or continuous electrical load** and during **unlimited run hours**. Depending on the **sites to supply** and the **availability of reliable utility**, the generating set manufacturer is responsible to define what power level he is able to supply to fulfil that requirement”
- “Prolonged operation at load **in parallel with a utility** is not permitted”



Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

ISO 8528-1 Data Centre Power Rating

Data Centre Power (ISO 8528-1)

- “Depending on the sites to supply and the **availability of reliable utility**, the generating set manufacturer is responsible to define what power level he is able to supply”
- “Prolonged operation at load **in parallel with a utility** is not permitted”

Because ISO 8528-1 Data Centre Power Rating references a reliable utility, it is not sufficient for Uptime Institute Tier III or Tier IV Certification

Rating	Emergency Standby Power	Prime Rated Power	Limited Time Prime	Continuous Operating Power	Data Centre Power
Load Profile	Variable	Variable	Constant	Constant	Variable OR Constant
Max Annual Run Hours	200	Unlimited	500	Unlimited	Unlimited

Data Center Continuous Ratings

- **Cummins Data Center Continuous (DCC) Ratings meet the Uptime Institute Tier III and Tier IV requirements**
- Data Center Continuous (DCC) Rating is defined as:
 - *The maximum power which the generator is capable of delivering continuously to a constant or varying electrical load for unlimited hours in a data center application.*



Model:	C3500 D6e
Frequency:	60 Hz
Fuel type:	Diesel
kW rating:	3500 standby 3000 prime 2750 continuous

Model:	C3500 D6e
Frequency:	60 Hz
Fuel type:	Diesel
kW rating:	3350 Data Center Continuous

Specification sheet

Diesel generator set QSK95 series engine

2750 kW - 3350 kW 60 Hz
Data Center Continuous
EPA Tier 2 emissions regulated

Description

Cummins Power Generation commercial generator sets are fully integrated power generation systems providing optimum performance, reliability and versatility for Data Center applications.

Features

Data Center Continuous (DCC) - Applicable for supplying power continuously to a constant or varying electrical load for unlimited hours in a data center application.

Uptime Compliant - Meets the requirement of a Tier III and IV data center site by being rated to run for unlimited hours of operation when loaded to 'N' demand for the engine generator set.

Derates

Altitude (air density) affects:

- Engine induction air
- Radiator cooling capacity
- Alternator cooling capacity

Ambient temperature affects:

- Engine induction
- Engine and radiator cooling capacity

Humidity

- Water particles in air



Performances Classes

There are four performance classes defined in ISO8528:

- **Class G1** - This applies to generating set applications where the connected loads are such that only *basic parameters of voltage and frequency* need to be specified.
 - E.g. General-purpose applications (lighting and other simple electrical loads).
- **Class G2** - This applies to generating set applications where its voltage characteristics are very *similar to those for the commercial public utility* electrical power system with which it operates. When load changes occur, *there can be temporary but acceptable deviations of voltage and frequency*.
 - E.g. Lighting systems, pumps, fans and hoists.
- **Class G3** - This applies to applications where *the connected equipment makes severe demands on the stability and level* of the frequency, voltage and waveform characteristics of the electrical power supplied by the generating set.
 - E.g. Telecommunications and thyristor-controlled loads. Both rectifier and thyristor-controlled loads can need special consideration with respect to their effect on generator-voltage waveform.
- **Class G4** - This applies to applications where the demands made on the stability and level of the frequency, voltage and waveform characteristics of the electrical power supplied by the generating set are exceptionally severe. *Agreement between manufacturer and customer*.
 - E.g. Data-processing equipment or computer systems.

Defining Acceptable Transient Performance



Requirements Overview

Class		Steady State	Frequency Droop	Load Rejection	Load Acceptance (Gas)	Recovery Band	Recovery Time
G1	Hz	2.5%	8%	18%	15% (25%)	3.5%	10 s
	V	±5%	---	35%	25%	10.0%	10 s
G2	Hz	1.5%	5%	12%	10% (20%)	2.0%	5 s
	V	±2.5%	---	25%	20%	5.0%	6 s
G3	Hz	0.5%	3%	10%	7% (15%)	2.0%	3 s
	V	±1%	---	20%	15%	2.0%	4 s

- 100% Load rejection (N/A <1kV from 2022)
- BMEP Loading
- 0% frequency droop (isochronous)
- G4 is set by Agreement between Manufacturer and Customer (AMC)

Note: this table only shows a high level overview. Consult ISO 8528-5 for the complete performance class operating limits.

BMEP

- Theoretical average pressure inside the engine cylinder
- It is a useful way of comparing relative engine performance

$$BMEP_{50\text{ Hz}} = 80 \times \frac{P}{V_d}$$

$$BMEP [kPa] = \frac{P \times n_r \times 1000}{V_d \times N}$$

P – Mechanical Power, kW

*n_r – number of crank revolutions
(2 for four stroke)*

V_d – cylinder volume, dm³

*N – Number of revolutions per second
(25 for 50 Hz and 30 for 60Hz)*

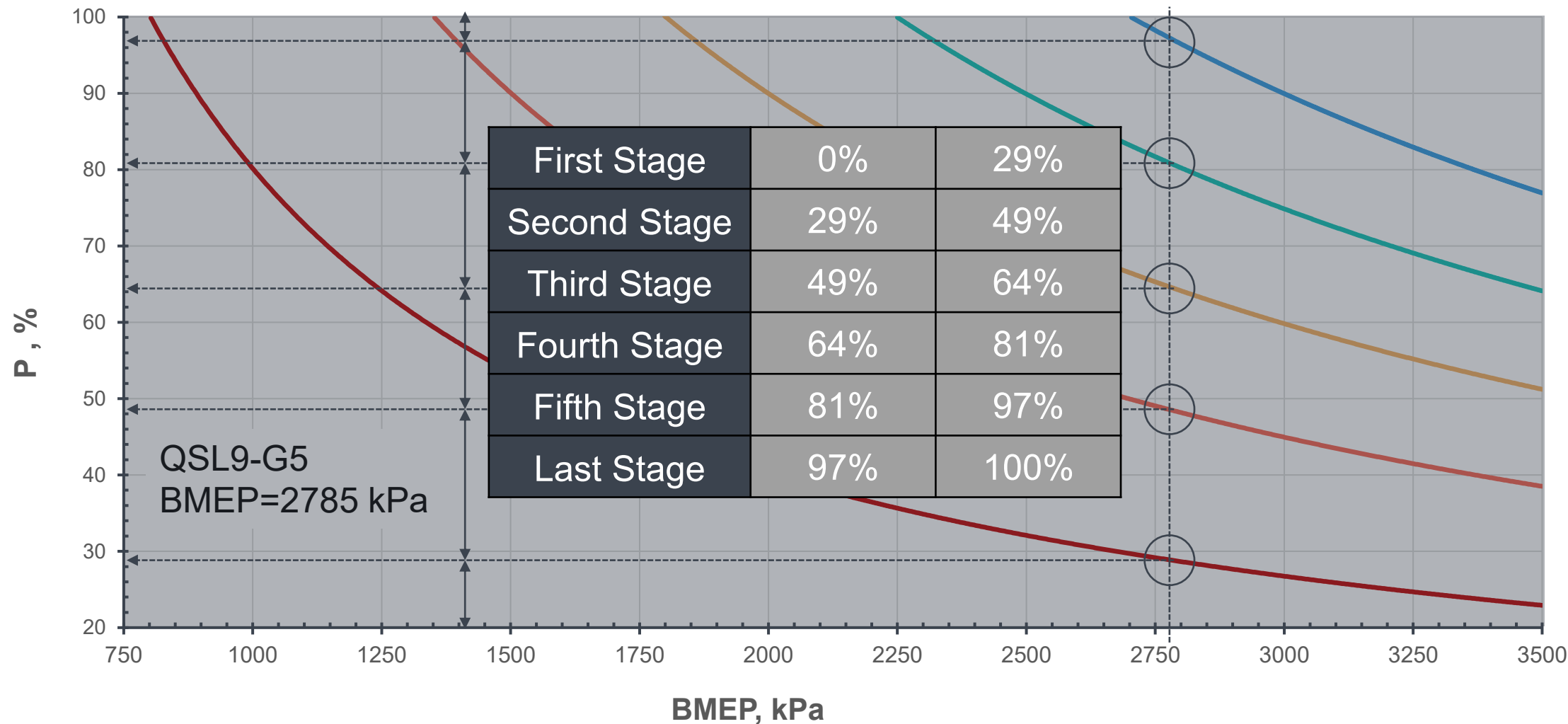
BMEP

- BMEP value for is available on datasheet
- Data for calculations can also be obtained in engine performance datasheet

QSZ13-G5 (469 kW @ 1500 rpm, 13 L)

		Standby Power		Prime Power	
Governed Engine Speed	RPM	1,800	1,500	1,800	1,500
Engine Idle Speed	RPM	650 - 750	650 - 750	650 - 750	650 - 750
Gross Engine PowerOutput	hp (kW)	671 (500)	630 (470)	586 (437)	551 (411)
Brake Mean EffectivePressure	psi (kPa)	372 (2,564)	419 (2,892)	325 (2,241)	367 (2,529)

Load Stages



Beyond ISO requirements

- Always understand customer requirements.
 - Meeting G3 performance class for 100% load rejection is often quite challenging... but is it really important for the application?
- Load profile has a significant impact on genset performance:
 - Application, power requirements of the connected load; load power factor; intermittent loads; effect of nonlinear loads, etc.
- G2 requirements are often acceptable within most commercial applications
- G4 performance class – Agreement between manufacturer & customer.

Q+A

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Self Access Internet Resource:

<https://powersuite.cummins.com>

Free access, requires only simple email registration.

Genset Sizing Tool and Data Library

Data & Specification sheets.

Drawings , BIM files



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